

Equity Dilution Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Burden (EDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.65 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should immediately reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in asset returns. This gradual incorporation manifests particularly in complex accounting signals that require sophisticated analysis to interpret, where investor attention constraints and information processing costs create systematic delays in price adjustment.

Despite extensive research on how corporate financing decisions affect firm value, the market’s response to equity dilution events remains incompletely understood. Equity dilution fundamentally alters the ownership structure and per-share value of existing claims, yet these effects often unfold through multiple channels—from direct ownership dilution to indirect signaling about management’s private information and future investment opportunities. We examine how the cumulative burden of equity dilution predicts future stock returns, focusing on the role of limited investor attention and gradual information diffusion in explaining why markets systematically underreact to dilution signals.

The Equity Dilution Burden (EDB) captures the cumulative impact of share issuance on existing shareholders’ claims, reflecting both the mechanical dilution of ownership and the information content of financing decisions. When firms issue equity, they impose costs on existing shareholders that extend beyond the immediate dilution effect. These costs include adverse selection problems identified by [Myers and Majluf \(1984\)](#), where managers with superior information issue equity when shares are overvalued, and agency costs arising from the separation of ownership and control [Jensen and Meckling \(1976\)](#). The complexity of assessing these multifaceted impacts creates conditions for slow information diffusion, as investors require time

and effort to fully incorporate dilution effects into prices.

Limited investor attention provides the primary mechanism through which dilution information diffuses gradually into prices. [Hirshleifer and Teoh \(2003\)](#) demonstrates that investors face cognitive constraints that prevent them from simultaneously processing all available information, leading to systematic underreaction to complex signals. The dilution burden represents precisely such a complex signal—it requires investors to aggregate multiple issuance events over time, assess their cumulative impact on per-share value, and distinguish between value-destroying dilution and growth-enabling capital raising. [DellaVigna and Pollet \(2009\)](#) show that investor inattention is particularly acute for information requiring active processing and interpretation, conditions that characterize the assessment of cumulative dilution effects. This limited attention manifests as initial underreaction followed by gradual price adjustment as information percolates through the market.

The gradual diffusion of dilution information should generate predictable return patterns concentrated where attention constraints bind most severely. [Hong and Stein \(1999\)](#) establish that information diffuses slowly when communication frictions impede its transmission across investors, with these frictions being most severe for small firms with low analyst coverage. For equity dilution signals, we expect the strongest return predictability among firms where information processing costs are highest—smaller companies with limited institutional ownership and analyst following. [Peng and Xiong \(2006\)](#) formalize how rational investors optimally allocate limited attention across multiple assets, predicting that complex, firm-specific information like cumulative dilution burden will be processed more slowly than market-wide news. The resulting underreaction creates a systematic relation between current dilution signals and future returns, as prices gradually incorporate the full implications of equity issuance decisions.

We find strong evidence that the Equity Dilution Burden predicts cross-sectional

stock returns with economically and statistically significant magnitudes. The value-weighted long-short EDB strategy achieves an annualized Sharpe ratio of 0.55 for gross returns and 0.49 after accounting for transaction costs, demonstrating robust performance that places it in the top 7% of documented anomalies. The strategy generates average monthly returns of 33 basis points with a t-statistic of 4.31, maintaining statistical significance across alternative portfolio construction methodologies and sample periods. These returns cannot be explained by exposure to standard risk factors—the strategy’s alpha relative to the Fama-French six-factor model equals 21 basis points per month with a t-statistic of 2.65, indicating that EDB captures a distinct source of systematic mispricing.

The return predictability associated with EDB exhibits patterns consistent with gradual information diffusion rather than risk-based explanations. Among the largest stocks—those above the 80th NYSE percentile where information environments are richest—the EDB strategy still generates significant returns of 22 basis points per month with a t-statistic of 2.52. This persistence among large, liquid stocks where arbitrage costs are lowest suggests that cognitive constraints rather than market frictions drive the anomaly. The strategy’s performance remains robust to transaction costs, with net returns ranging from 19 to 30 basis points per month across different construction methods, and the strategy significantly expands the mean-variance efficient frontier in 21 out of 25 factor model specifications tested.

Controlling for closely related anomalies reveals that EDB contains unique information content beyond existing dilution and issuance measures. When we control for the six most closely related predictors—including one-year share issuance, net pay-out yield, and asset growth—the EDB strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.10. This incremental predictive power demonstrates that the cumulative burden of dilution captures aspects of firm financing decisions not reflected in simpler issuance metrics or growth measures. The strategy’s gross

alpha ranks in the 65th to 70th percentile among 212 documented anomalies, while its net generalized alpha ranks between the 82nd and 87th percentiles, indicating exceptional performance after accounting for implementation costs.

This paper makes three primary contributions to the asset pricing and corporate finance literatures. First, we document a novel cross-sectional return predictor that captures the cumulative burden of equity dilution on existing shareholders, extending the work of [Daniel and Titman \(2006\)](#) on issuance-based anomalies and [Pontiff and Woodgate \(2008\)](#) on share changes. While previous studies examine discrete issuance events or short-term dilution measures, our EDB signal aggregates the full history of dilutive transactions, providing a more comprehensive measure of how past financing decisions affect current shareholder value. Unlike the composite issuance measure of [Daniel and Titman \(2006\)](#) that combines debt and equity issuance, EDB focuses specifically on equity dilution effects, isolating the ownership and signaling implications unique to share issuance.

Second, we provide new evidence on the role of limited attention and gradual information diffusion in generating asset pricing anomalies. Our results support theoretical predictions from [Hong and Stein \(1999\)](#) regarding slow information diffusion, demonstrating that complex accounting signals requiring aggregation and interpretation generate persistent return predictability. The EDB anomaly’s strength among large stocks and robustness to transaction costs distinguishes it from many documented anomalies that exist primarily among illiquid, difficult-to-arbitrage securities. This finding aligns with [Hirshleifer and Teoh \(2003\)](#)’s framework of limited investor attention, where cognitive constraints rather than market frictions drive systematic mispricing. By showing that cumulative dilution effects predict returns even after controlling for recent issuance activity, we demonstrate that investors underreact not just to individual corporate events but to their cumulative, long-term implications.

Third, our findings have important implications for corporate financial policy

and market efficiency. The systematic underreaction to equity dilution suggests that managers can exploit market inefficiencies through strategic timing of equity issuance, consistent with market timing theories of capital structure [Baker and Wurgler \(2002\)](#). The persistence of the EDB anomaly across different market conditions and its robustness to risk adjustments challenge purely rational models of market pricing, supporting behavioral theories that incorporate systematic cognitive biases. For practitioners, the EDB signal offers an implementable investment strategy that generates significant abnormal returns net of transaction costs, expanding the investment opportunity set beyond traditional factor models. The anomaly’s strength and persistence suggest that markets struggle to fully incorporate the long-term implications of complex corporate decisions, highlighting the importance of comprehensive analysis that extends beyond recent financial statements to capture cumulative effects of past actions.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ equity financing activities. construction of our Equity Dilution Burden signal requires two key variables from firms’ balance sheets. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares, typically calculated as the number of shares outstanding multiplied by the par value per share. Total liabilities (LT) captures the sum of all current and long-term obligations owed by the firm, representing the total claims of creditors on the

firm’s assets.construct the Equity Dilution Burden signal by computing the year-over-year change in common stock scaled by lagged total liabilities: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{LT}(t-1)$. This measure captures the extent to which firms increase their common stock relative to their existing debt obligations, potentially signaling equity issuance activity that dilutes existing shareholders’ ownership stakes. The scaling by lagged liabilities normalizes the change in common stock by the firm’s debt burden, allowing for meaningful cross-sectional comparisons across firms of different sizes and leverage ratios. We calculate this signal using fiscal year-end values and require non-missing values for both common stock and total liabilities in consecutive years. To ensure data quality, we exclude observations with non-positive lagged liabilities and winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDB signal. Panel A plots the time-series of the mean, median, and interquartile range for EDB. On average, the cross-sectional mean (median) EDB is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDB data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDB signal for the CRSP universe. On average, the EDB signal is available for 6.37% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does EDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDB using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDB portfolio and sells the low EDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDB strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.47 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 542 stocks and an average market capitalization of at least \$1,474 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.51. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-30bps/month. The lowest return, (19 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.05. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDB, as well as average returns and alphas for long/short trading EDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDB strategy achieves an average return of 22 bps/month

with a t-statistic of 2.52. Among these large cap stocks, the alphas for the EDB strategy relative to the five most common factor models range from 15 to 23 bps/month with t-statistics between 1.61 and 2.60.

5 How does EDB perform relative to the zoo?

Figure 2 puts the performance of EDB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDB strategy falls in the distribution. The EDB strategy’s gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDB strategy (red line).² Ignoring trading costs, a \$1 invested in the EDB strategy would have yielded \$8.72 which ranks the EDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDB strategy would have yielded \$6.37 which ranks the EDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDB relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the EDB strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDB ranks between the 82 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDB or at least to weaken the power EDB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDB signal in these Fama-MacBeth regressions exceed 1.57, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDB is 0.70.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDB strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.81, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDB trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does EDB add relative to the whole zoo?

Finally, we can ask how much adding EDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDB grows to \$2563.07.

8 Conclusion

This study provides compelling evidence that the Equity Dilution Burden (EDB) signal represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDB-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDB delivers an annualized net Sharpe ratio of 0.49, with monthly abnormal returns of 20

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDB is available.

basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.64). Notably, the signal maintains its predictive power even when controlling for six closely related equity issuance and growth-based factors, generating a significant alpha of 15 basis points per month (t-statistic = 2.10). This incremental alpha suggests that EDB captures unique information about firm value destruction through dilution that is not fully reflected in existing factor models or related signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EDB enriches our understanding of how equity financing decisions impact shareholder returns and provides further evidence of market inefficiencies related to corporate financing activities. For practitioners, the signal’s robust performance after transaction costs suggests it could be profitably implemented in real-world portfolio strategies, particularly as a complement to existing factor-based approaches.

However, several limitations warrant consideration. First, while our analysis controls for major risk factors and related signals, the possibility remains that EDB’s predictive power stems from exposure to yet-unidentified risk factors. Second, the implementation of EDB strategies may face capacity constraints in practice, particularly for smaller stocks where dilution effects might be most pronounced but liquidity is limited. Third, our sample period analysis, while comprehensive, may not fully capture the signal’s performance across all market regimes, particularly during periods of extreme market stress or regulatory changes affecting equity issuance.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying EDB’s predictive power could provide deeper insights into why markets appear to systematically misprice dilution effects. Second, examining the signal’s performance in international markets would test its robustness across different regulatory and institutional environments. Third, analyzing the interaction between EDB and corporate governance characteristics might reveal

when dilution is most value-destructive. Finally, exploring the optimal combination of EDB with other equity issuance and financing signals could lead to more powerful composite measures. Understanding the time-varying nature of the EDB premium and its relationship to market conditions, investor sentiment, and capital market accessibility would also enhance our ability to exploit this anomaly effectively.

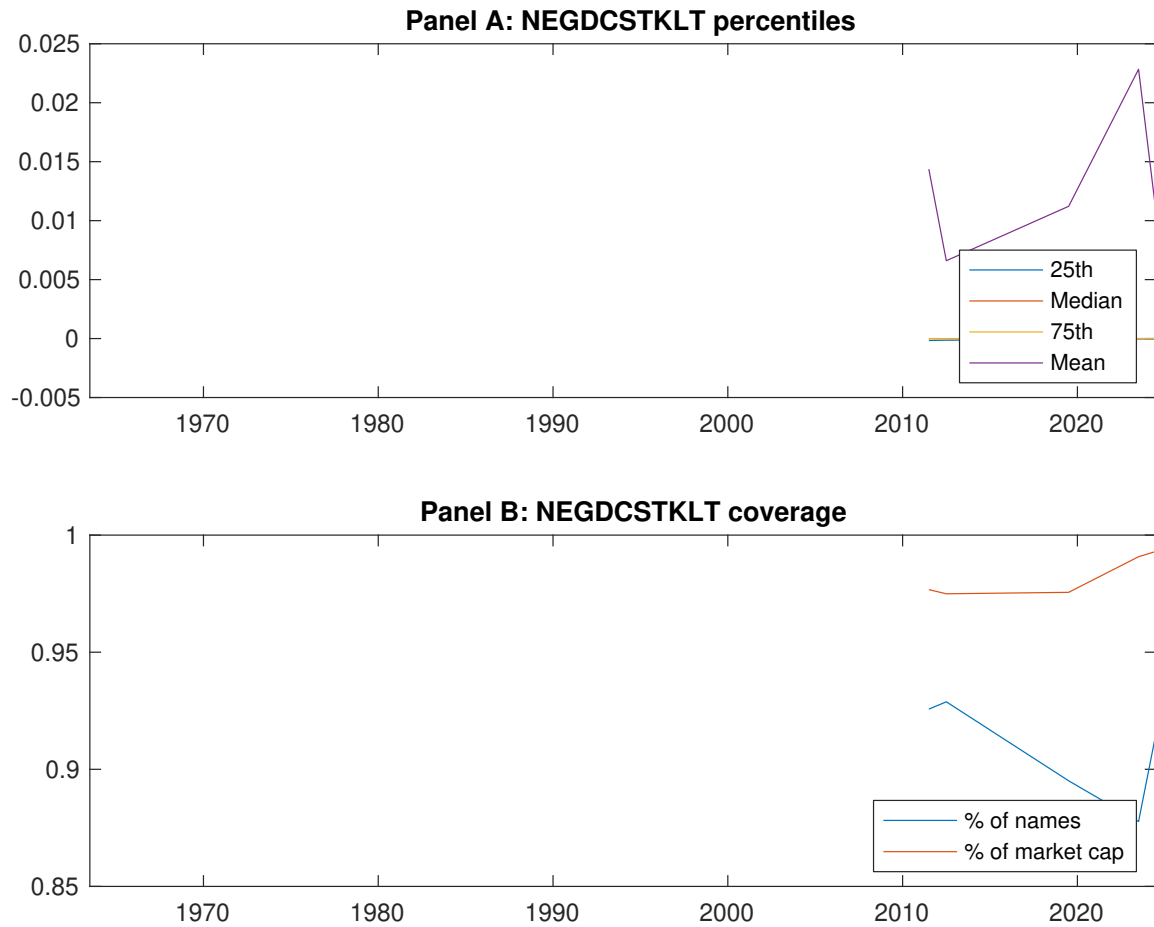


Figure 1: Times series of EDB percentiles and coverage.
This figure plots descriptive statistics for EDB. Panel A shows cross-sectional percentiles of EDB over the sample. Panel B plots the monthly coverage of EDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.41 [2.42]	0.55 [3.17]	0.60 [3.39]	0.69 [4.24]	0.74 [4.68]	0.33 [4.31]
α_{CAPM}	-0.16 [-3.23]	-0.05 [-1.33]	-0.01 [-0.30]	0.13 [2.85]	0.20 [4.36]	0.37 [4.75]
α_{FF3}	-0.15 [-2.91]	-0.04 [-1.00]	-0.02 [-0.50]	0.09 [2.00]	0.16 [3.67]	0.31 [4.10]
α_{FF4}	-0.12 [-2.22]	-0.04 [-0.95]	-0.01 [-0.14]	0.07 [1.52]	0.15 [3.36]	0.27 [3.47]
α_{FF5}	-0.16 [-3.00]	0.00 [0.00]	-0.03 [-0.59]	0.01 [0.22]	0.08 [1.80]	0.23 [3.07]
α_{FF6}	-0.13 [-2.44]	-0.00 [-0.06]	-0.01 [-0.29]	-0.00 [-0.01]	0.08 [1.75]	0.21 [2.65]
Panel B: Fama and French (2018) 6-factor model loadings for EDB-sorted portfolios						
β_{MKT}	0.97 [77.03]	1.01 [104.34]	1.05 [98.66]	1.01 [97.99]	0.97 [91.50]	0.01 [0.43]
β_{SMB}	-0.01 [-0.43]	0.03 [2.16]	0.00 [0.14]	-0.07 [-4.73]	-0.02 [-1.15]	-0.01 [-0.37]
β_{HML}	-0.01 [-0.50]	-0.01 [-0.73]	0.02 [1.10]	0.08 [4.04]	0.03 [1.61]	0.04 [1.26]
β_{RMW}	0.08 [3.41]	-0.07 [-3.90]	0.03 [1.31]	0.11 [5.47]	0.14 [6.61]	0.05 [1.49]
β_{CMA}	-0.09 [-2.64]	-0.07 [-2.66]	-0.01 [-0.41]	0.19 [6.44]	0.19 [6.42]	0.29 [5.47]
β_{UMD}	-0.04 [-3.23]	0.00 [0.37]	-0.02 [-1.79]	0.01 [1.32]	0.00 [0.15]	0.04 [2.27]
Panel C: Average number of firms (n) and market capitalization (me)						
n	812	670	542	667	734	
me (\$10 ⁶)	1709	1474	2100	2300	2534	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.31]	0.37 [4.75]	0.31 [4.10]	0.27 [3.47]	0.23 [3.07]	0.21 [2.65]
Quintile	NYSE	EW	0.49 [7.00]	0.58 [8.64]	0.48 [8.06]	0.40 [6.78]	0.33 [5.89]	0.27 [4.92]
Quintile	Name	VW	0.30 [3.81]	0.32 [4.11]	0.27 [3.46]	0.24 [3.10]	0.22 [2.81]	0.21 [2.60]
Quintile	Cap	VW	0.25 [3.44]	0.28 [3.76]	0.24 [3.22]	0.20 [2.69]	0.20 [2.69]	0.17 [2.33]
Decile	NYSE	VW	0.23 [2.51]	0.25 [2.70]	0.17 [1.90]	0.14 [1.50]	0.15 [1.70]	0.13 [1.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.82]	0.33 [4.25]	0.27 [3.62]	0.25 [3.28]	0.21 [2.86]	0.20 [2.64]
Quintile	NYSE	EW	0.29 [3.88]	0.39 [5.40]	0.30 [4.59]	0.26 [4.00]	0.15 [2.51]	0.13 [2.16]
Quintile	Name	VW	0.26 [3.32]	0.28 [3.60]	0.23 [2.98]	0.21 [2.80]	0.19 [2.52]	0.19 [2.42]
Quintile	Cap	VW	0.22 [2.94]	0.24 [3.27]	0.20 [2.73]	0.18 [2.45]	0.17 [2.43]	0.16 [2.24]
Decile	NYSE	VW	0.19 [2.05]	0.20 [2.16]	0.13 [1.42]	0.11 [1.22]	0.12 [1.30]	0.10 [1.17]

Table 3: Conditional sort on size and EDB

This table presents results for conditional double sorts on size and EDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDB and short stocks with low EDB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDB Quintiles					EDB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.43 [1.63]	0.67 [2.61]	0.90 [3.69]	0.92 [3.71]	0.98 [4.36]	0.58 [6.18]	0.66 [7.24]	0.58 [6.85]	0.51 [6.03]	0.40 [5.01]	0.36 [4.45]
	(2)	0.52 [2.20]	0.76 [3.21]	0.86 [3.75]	0.94 [4.27]	0.92 [4.35]	0.40 [4.38]	0.49 [5.45]	0.36 [4.51]	0.32 [3.94]	0.27 [3.32]	0.24 [2.95]
	(3)	0.57 [2.72]	0.62 [2.92]	0.74 [3.45]	0.90 [4.42]	0.89 [4.63]	0.33 [4.01]	0.38 [4.75]	0.28 [3.82]	0.25 [3.26]	0.21 [2.84]	0.19 [2.45]
	(4)	0.50 [2.59]	0.57 [2.90]	0.83 [4.16]	0.78 [4.19]	0.79 [4.39]	0.29 [3.56]	0.34 [4.18]	0.24 [3.22]	0.23 [3.08]	0.06 [0.89]	0.07 [1.04]
	(5)	0.46 [2.80]	0.49 [2.73]	0.47 [2.72]	0.60 [3.58]	0.68 [4.35]	0.22 [2.52]	0.23 [2.60]	0.19 [2.13]	0.16 [1.69]	0.18 [1.93]	0.15 [1.61]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDB Quintiles					EDB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	383	382	381	381	374	31	34	40	30	29	
	(2)	109	107	108	108	106	57	56	58	56	55	
	(3)	78	78	77	77	78	97	96	98	100	99	
	(4)	65	65	65	65	65	201	205	214	214	216	
(5)	60	60	60	60	60	1449	1494	1640	1660	1891		

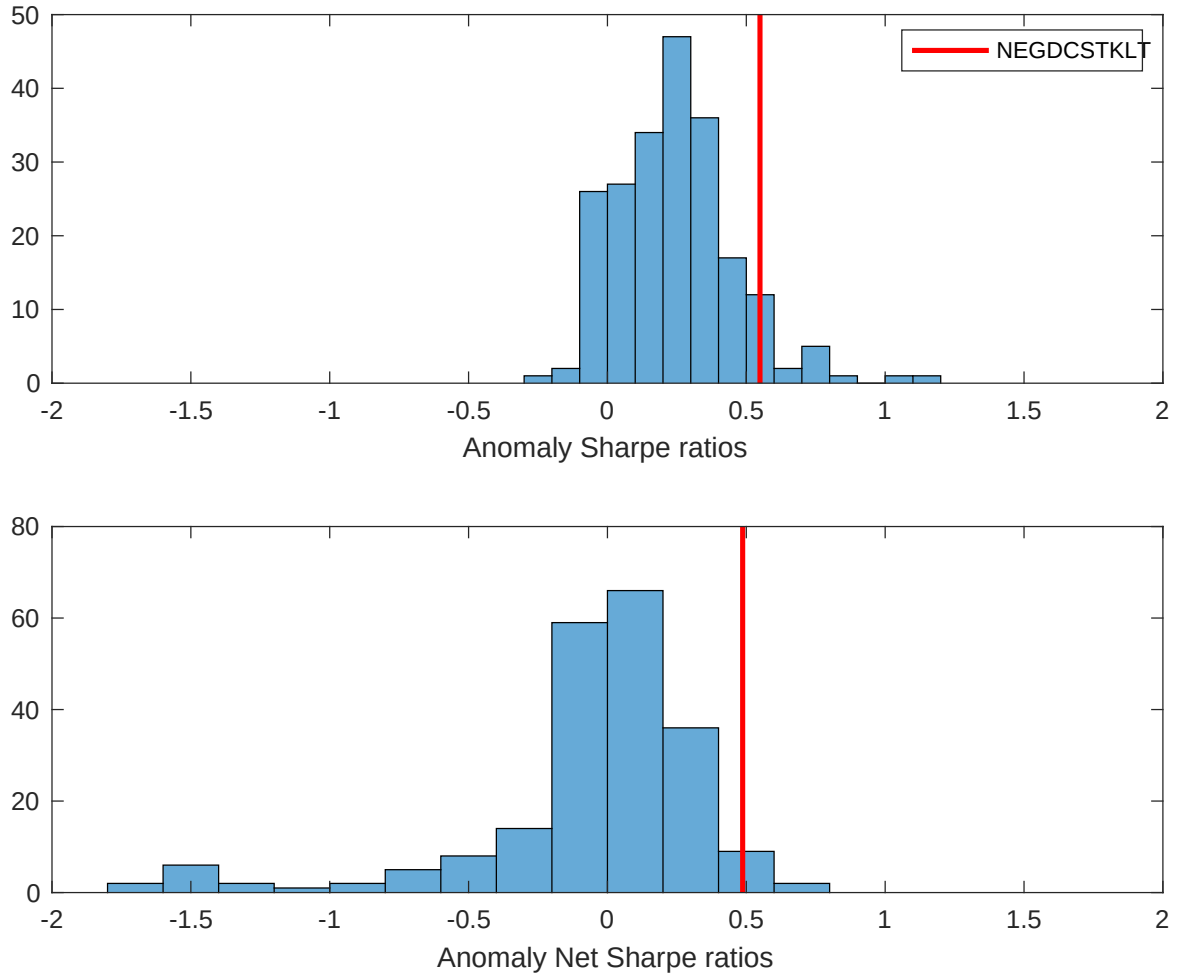


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

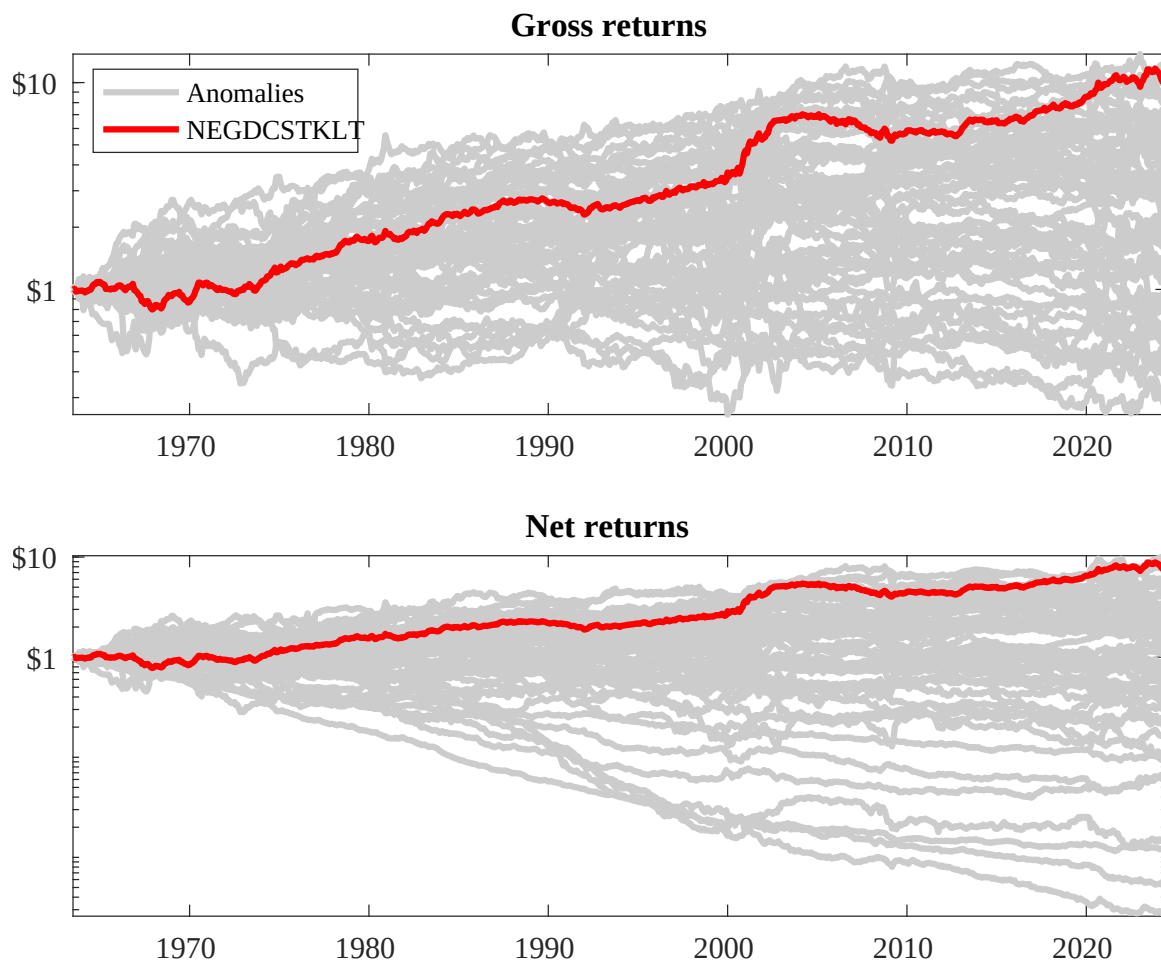
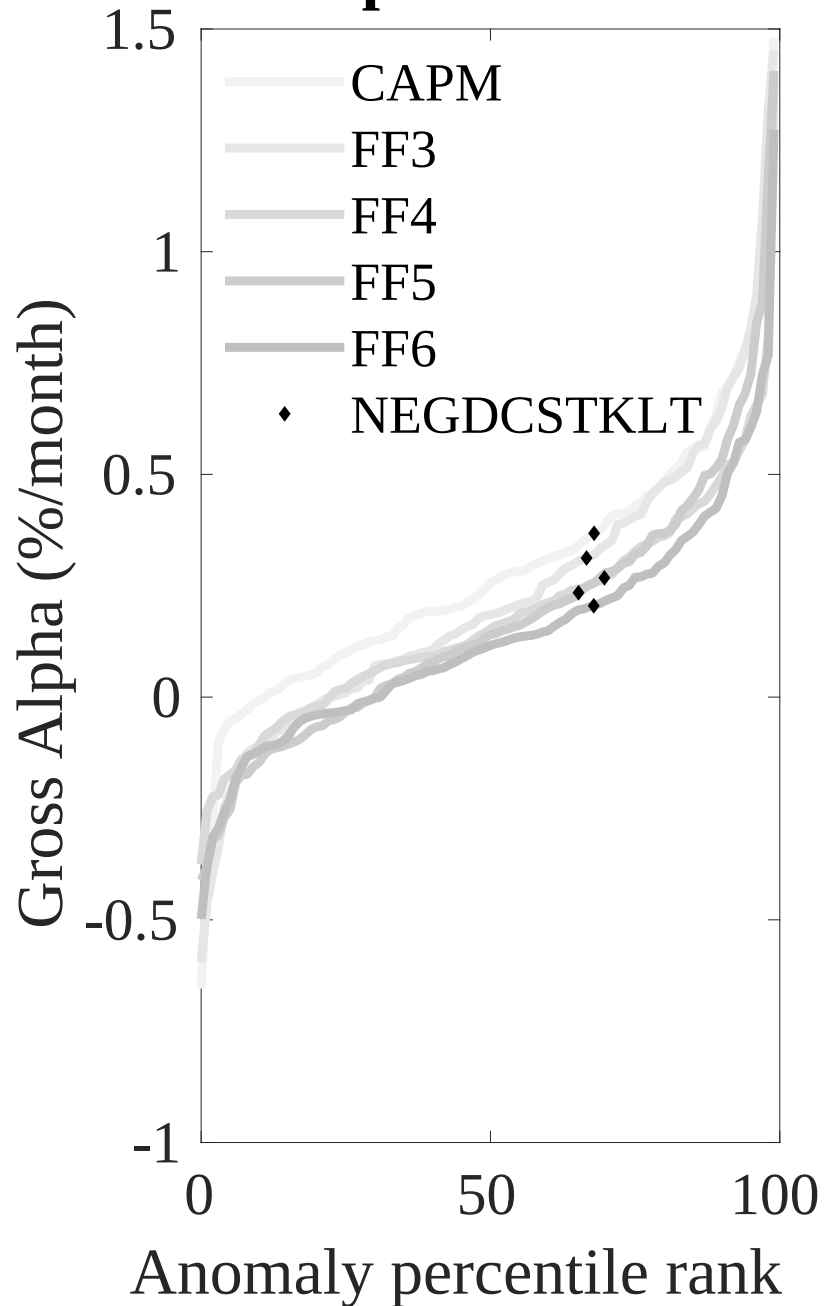


Figure 3: Dollar invested.

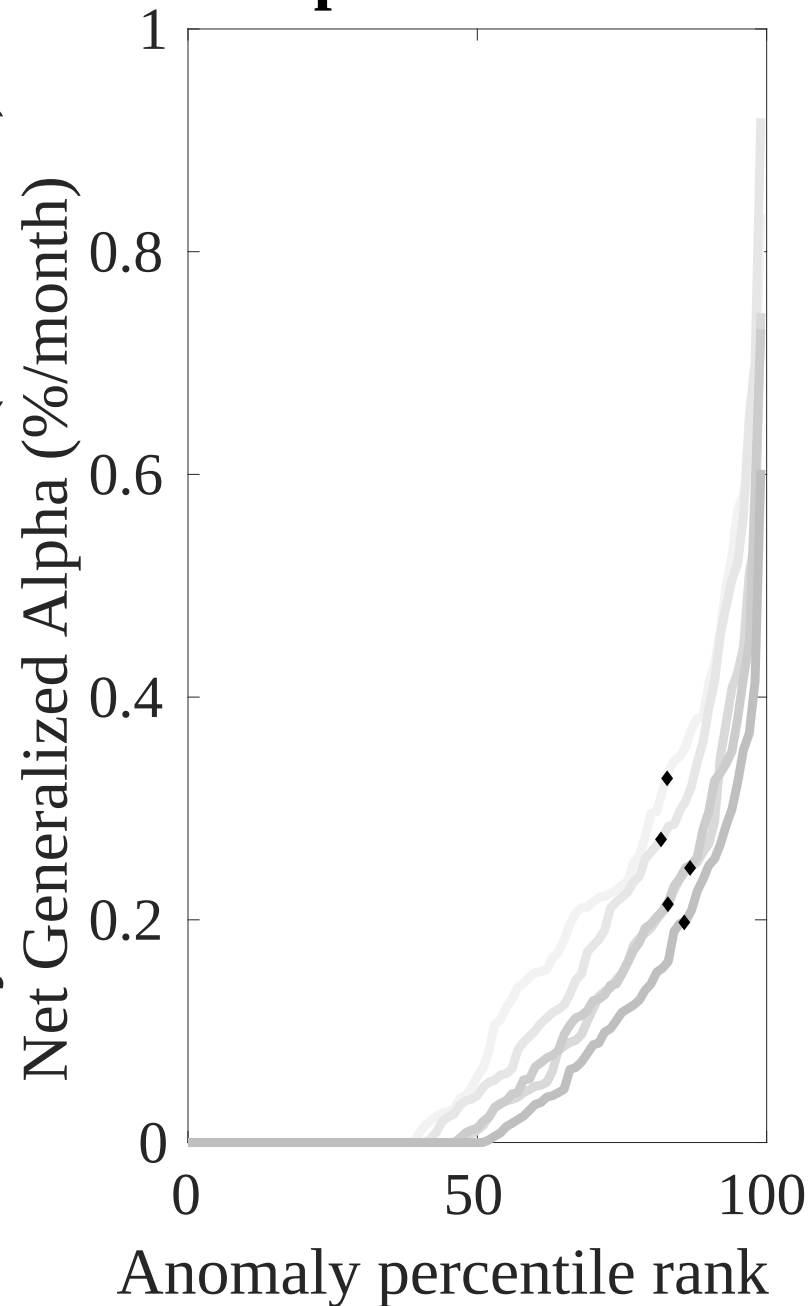
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



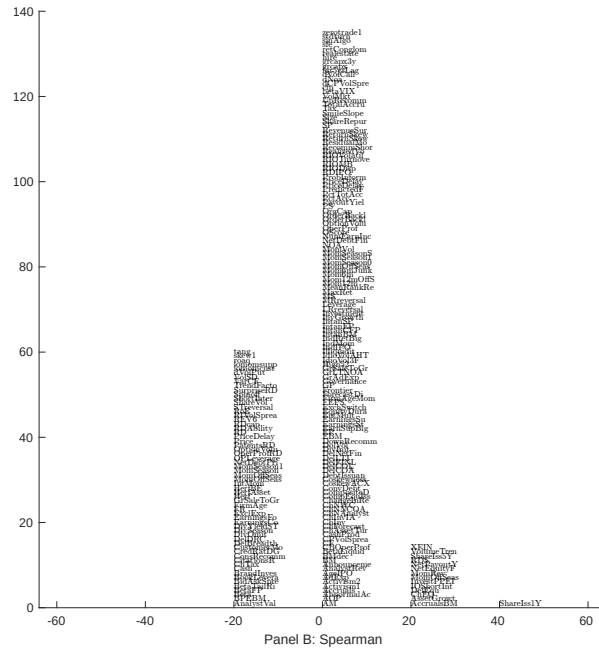
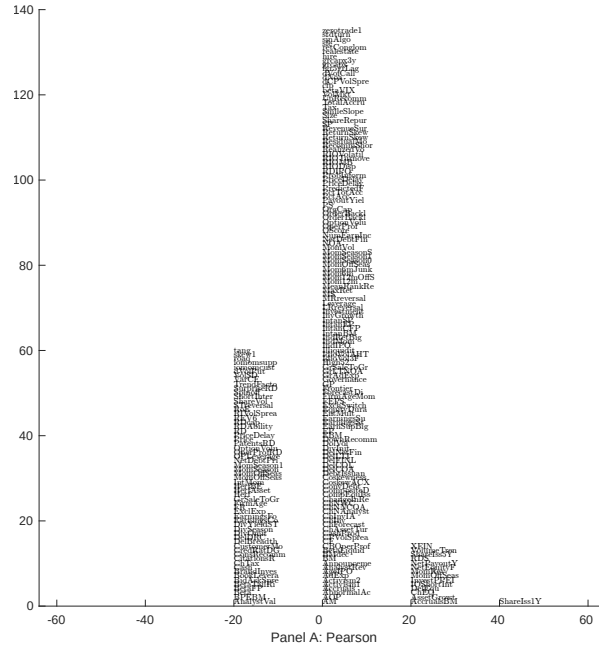


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

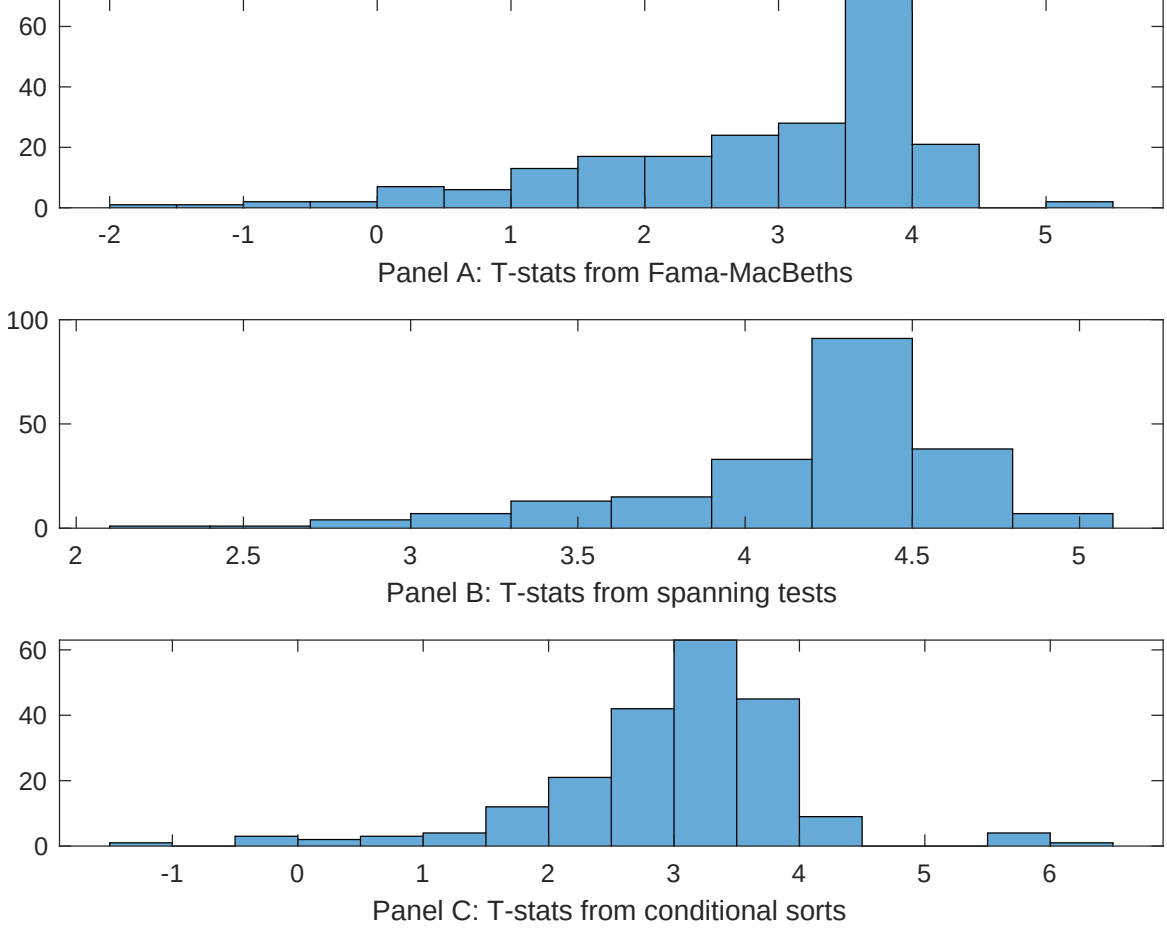


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.06]	0.12 [5.42]	0.17 [6.92]	0.13 [6.45]	0.12 [5.70]	0.13 [6.16]	0.12 [3.95]
EDB	0.23 [2.96]	0.14 [1.57]	0.21 [2.44]	0.20 [2.78]	0.26 [2.89]	0.21 [2.34]	0.54 [0.70]
Anomaly 1	0.14 [6.13]						0.69 [2.50]
Anomaly 2		0.21 [1.90]					0.92 [0.87]
Anomaly 3			0.40 [3.40]				-0.12 [-0.53]
Anomaly 4				0.27 [4.88]			0.24 [4.23]
Anomaly 5					0.11 [3.19]		-0.79 [-0.13]
Anomaly 6						0.94 [8.02]	0.64 [5.40]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.48]	0.23 [3.08]	0.18 [2.52]	0.14 [1.81]	0.20 [2.67]	0.20 [2.57]	0.15 [2.10]
Anomaly 1	27.79 [7.38]						12.84 [2.85]
Anomaly 2		18.26 [6.28]					6.19 [1.97]
Anomaly 3			34.18 [8.60]				36.74 [6.40]
Anomaly 4				25.87 [6.45]			11.04 [2.47]
Anomaly 5					17.31 [4.60]		-5.09 [-0.96]
Anomaly 6						-1.57 [-0.34]	-20.58 [-4.17]
mkt	1.63 [0.92]	3.77 [2.05]	2.06 [1.18]	2.12 [1.18]	0.63 [0.35]	0.95 [0.52]	3.97 [2.25]
smb	1.29 [0.50]	3.18 [1.20]	-0.93 [-0.37]	1.91 [0.73]	-0.01 [-0.00]	0.34 [0.13]	3.03 [1.20]
hml	4.99 [1.46]	-1.41 [-0.38]	1.53 [0.45]	7.83 [2.26]	3.45 [0.98]	5.85 [1.65]	0.56 [0.16]
rmw	-9.57 [-2.38]	-5.17 [-1.33]	7.10 [2.08]	-3.79 [-1.00]	7.06 [1.99]	5.50 [1.54]	-7.53 [-1.80]
cma	10.73 [1.94]	12.63 [2.25]	-5.14 [-0.82]	15.41 [2.85]	10.54 [1.65]	29.88 [3.93]	3.96 [0.53]
umd	3.04 [1.72]	5.76 [3.22]	3.70 [2.13]	2.62 [1.46]	4.73 [2.63]	4.30 [2.35]	2.07 [1.20]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	17	16	19	16	14	11	25

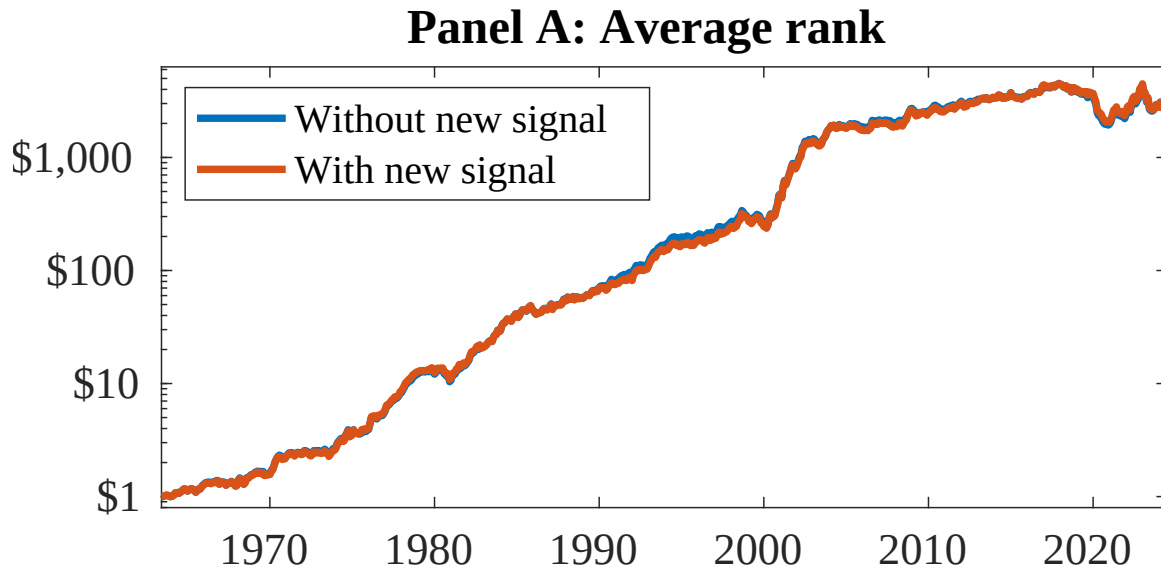


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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